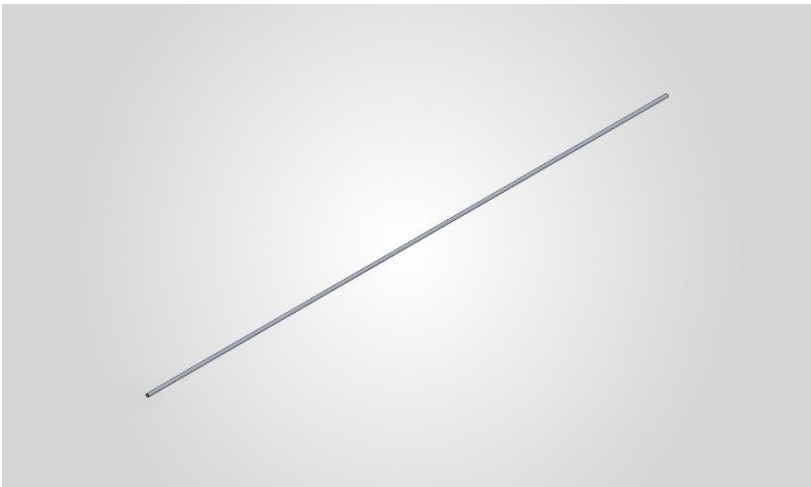




Description
No Data

Simulation of A3 FEA beam

Date: Friday, September 4, 2026
Designer: Solidworks
Study name: SimulationXpress Study
Analysis type: Static

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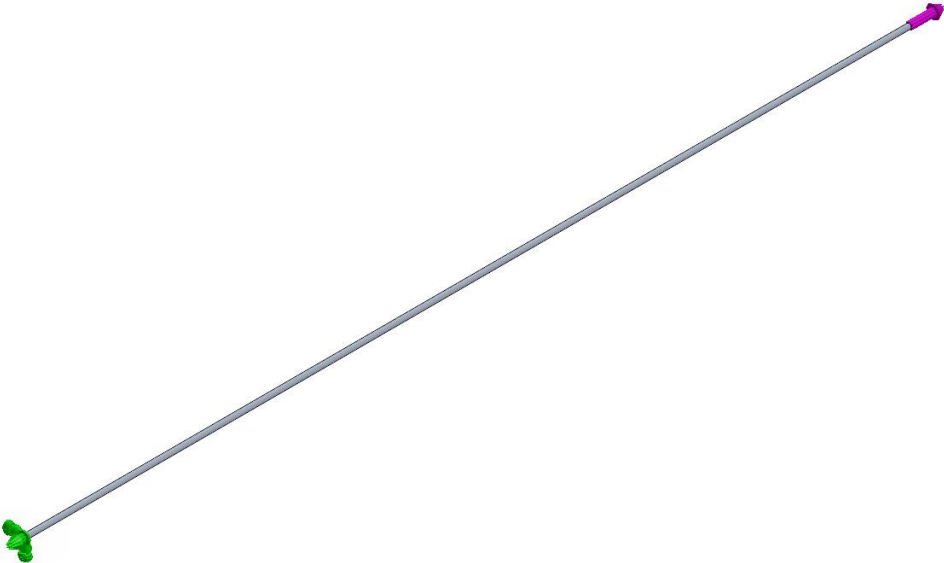
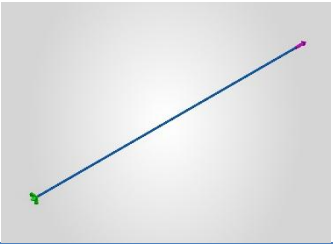
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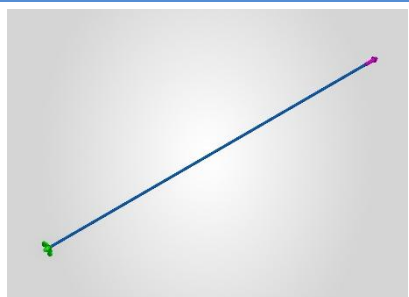
Assumptions

Model Information

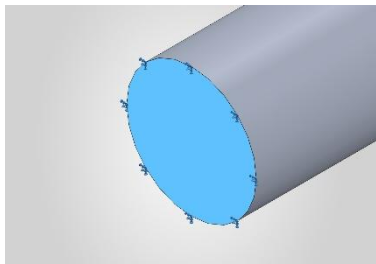
			
Model name: A3 FEA beam Current Configuration: Default			
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	Mass:6.14067 kg Volume:0.00227432 m^3 Density:2,700 kg/m^3 Weight:60.1786 N	C:\Users\twins\Documents \SOLIDWORKS Downloads\A3 FEA beam.SLDPRT Sep 3 18:14:19 2026

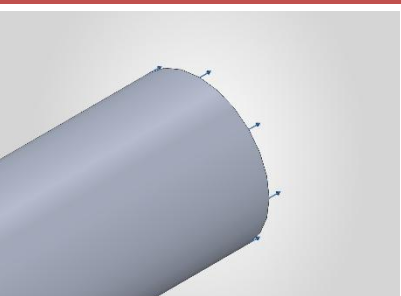


Material Properties

Model Reference	Properties	Components
	Name: 3.0205 (EN-AW 1200) Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 2.5×10^7 N/m ² Tensile strength: 7.5×10^7 N/m ²	SolidBody 1(Boss-Extrude1)(A3 FEA beam)

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-1 (:Per item: -400 lbf:)		Entities: 1 face(s) Type: Apply normal force Value: -400 lbf



Mesh information

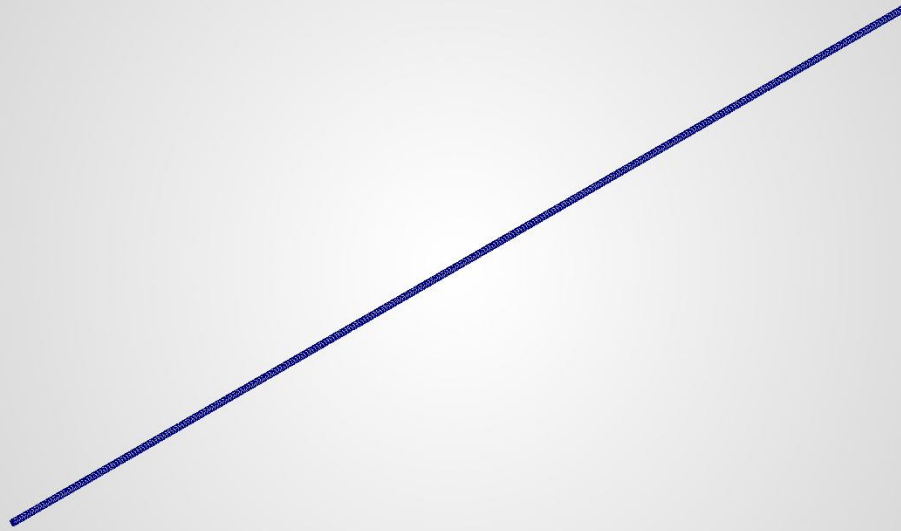
Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	16 Points
Element Size	0.517851 in
Tolerance	0.0258926 in
Mesh Quality	High

Mesh information - Details

Total Nodes	13122
Total Elements	6225
Maximum Aspect Ratio	3.6589
% of elements with Aspect Ratio < 3	99.9
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:01
Computer name:	



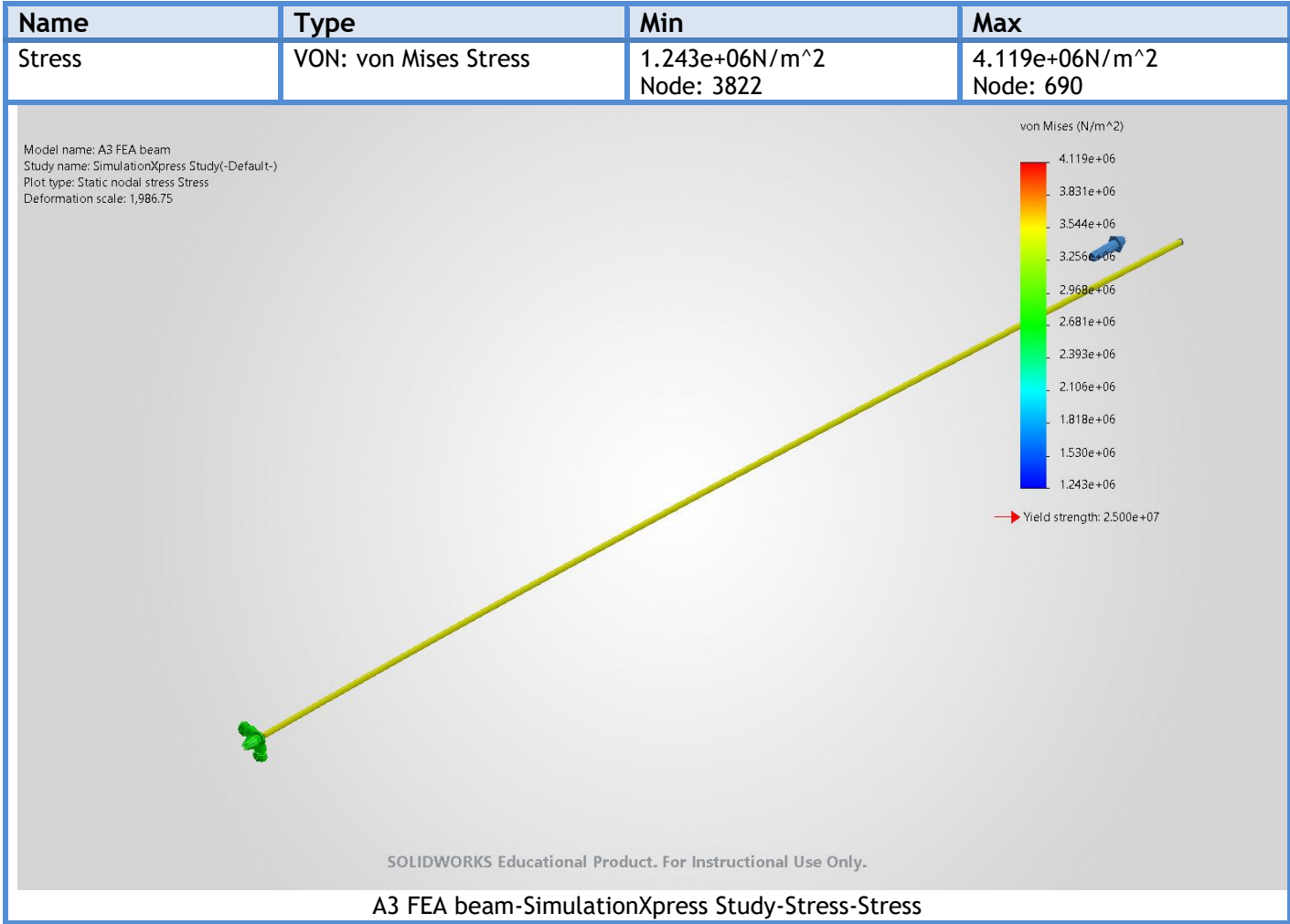
Model name: A3 FEA beam
Study name: SimulationXpress Study(-Default-)
Mesh type: Solid Mesh



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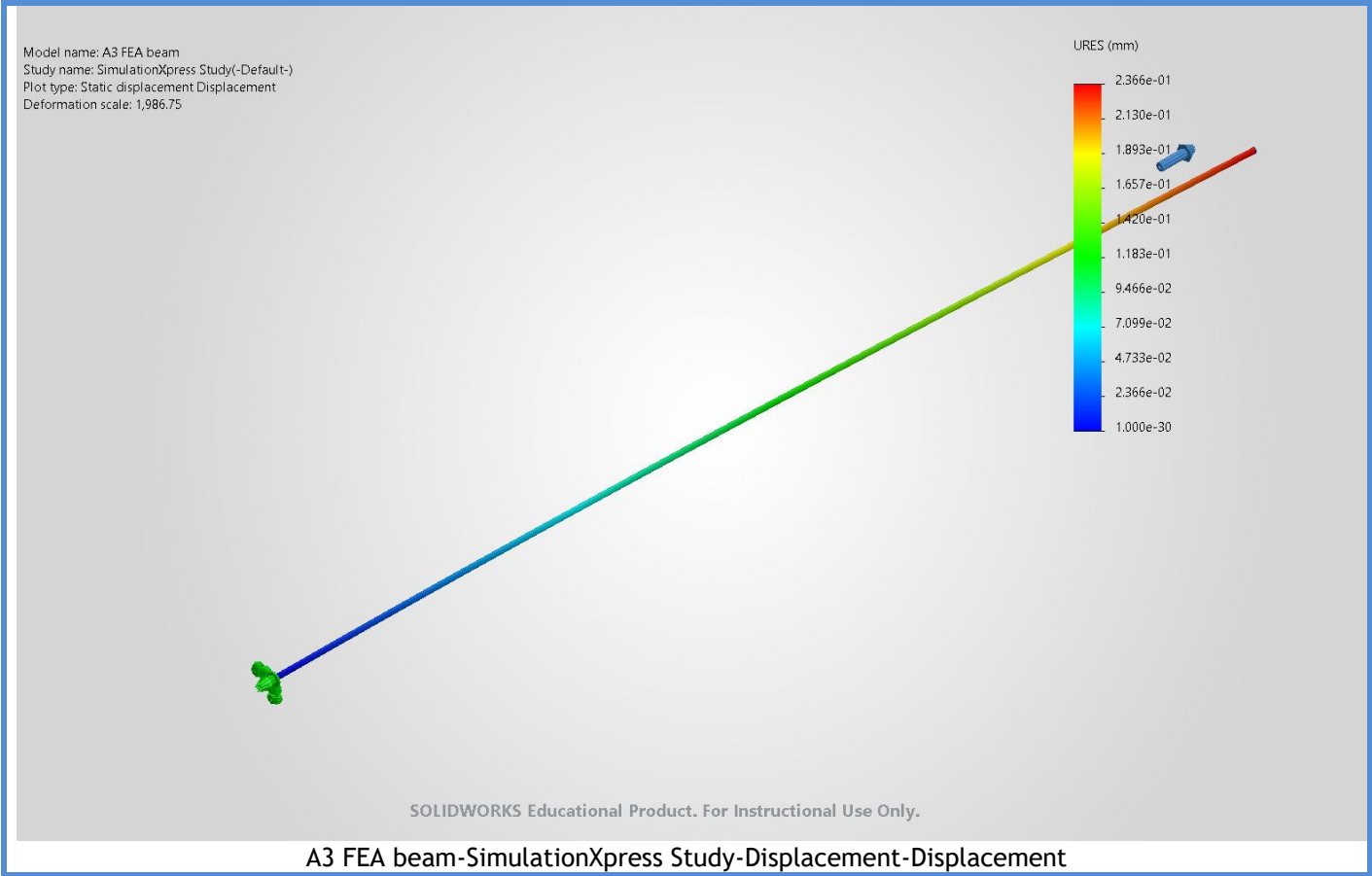


Study Results



Name	Type	Min	Max
Displacement	URES: Resultant Displacement	0.000e+00mm Node: 1	2.366e-01mm Node: 1035

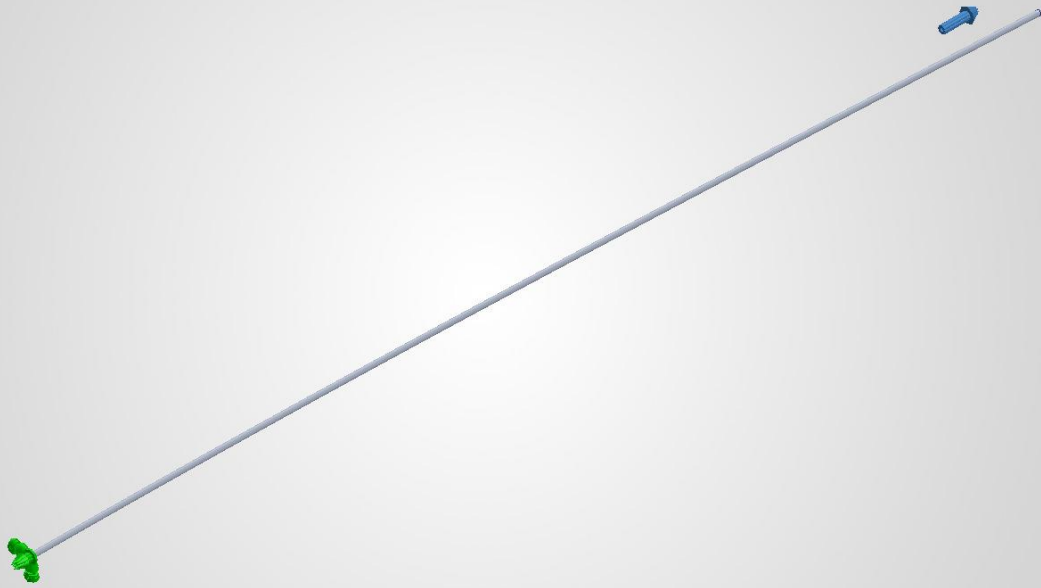




Name	Type
Deformation	Deformed shape



Model name: A3 FEA beam
Study name: SimulationXpress Study(-Default-)
Plot type: Deformed shape Deformation
Deformation scale: 1,986.75



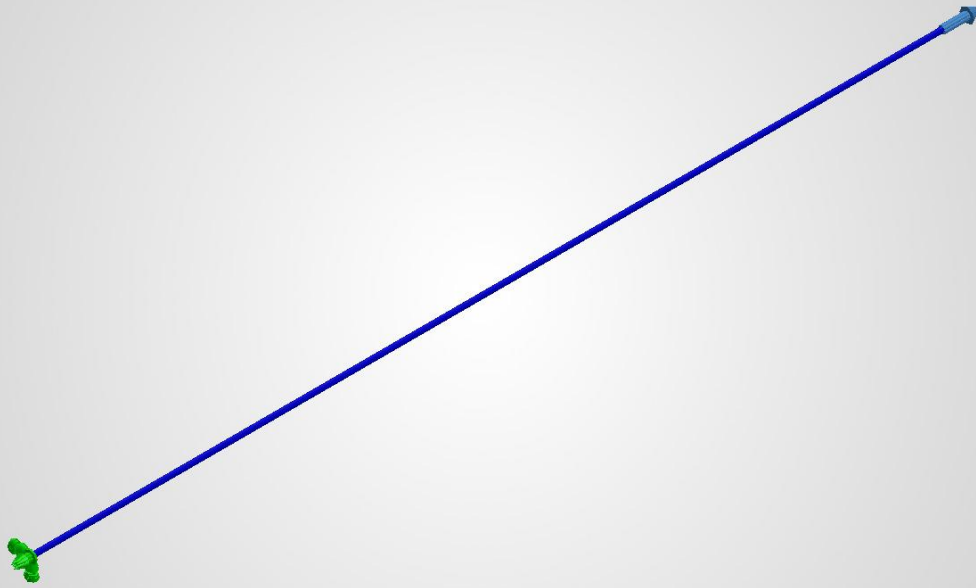
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A3 FEA beam-SimulationXpress Study-Displacement-Deformation

Name	Type	Min	Max
Factor of Safety	Max von Mises Stress	6.070e+00 Node: 690	2.012e+01 Node: 3822



Model name: A3 FEA beam
Study name: SimulationXpress Study(-Default-)
Plot type: Factor of Safety Factor of Safety
Criterion: Max von Mises Stress
Red < FOS = 1 < Blue



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A3 FEA beam-SimulationXpress Study-Factor of Safety-Factor of Safety

Conclusion



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Analyzed with SOLIDWORKS Simulation

Simulation of A3 FEA beam